Unit - I

Natural Hazards and Disaster Management

Introduction:

India is one of the most vulnerable countries in the world to a excess of natural disasters.

→ While floods and droughts are recurrent disasters in a few states, tropical cyclones hit most of the east coast, which is more than 1000 Km long on regular basis.

→ About 57% area of the country is highly seismic in nature.

→ India's vast coastline, especially the entire eastern coast from West Bengal to Tamilnadu and part of the western coast of Gujarat and Maharashtra, is highly vulnerable to tropical cyclones

→ India's long Himalayan mountain range in north, is geologically a young range of mountain chains. Every year we face several hundred landslides from Jammu & Kashmir to the north eastern states & there is loss of life and property.

→ Earthquakes, Tsunamis, Volcanoes and forest fires have brought fear and a sense of insecurity in the minds of people.

→ Disaster management is a process of effectively preparing for and responding to disasters.

→ It involves strategically organizing resources to reduce the harm that disasters cause.

→ It also involves a systematic approach to managing the responsibilities of disaster prevention, preparedness, response and recovery.

→ Hence the disaster management has a greater importance

Disaster (Definition):

A serious disruption of the functioning of a community or a society causing widespread human, material, economic (or) environmental losses that exceed the ability of the affected community or society to cope using its own resources.

→ The term "DISASTER" owes its origin to french word "Disastre" – combination of two words 1) "Des" meaning "Bad"

2) "Aster" meaning "Star"

Hence the term Disaster refers to "Bad (or) Evil star"

Types of Disaster's

1) Natural Disasters
2) Man made Disasters
3) Incidences of Mass Trauma

Disasters

- Natural Disasters
  - Meteorological (Floods, cyclone, Hurrican, Typhoon, Snowstorm, Thunderstrom, Droughts, Hot/cold waves)
  - Geological (Earthquake, Volcanic Eruptions, Landslides, Tsunamis, Debris/Mud flows)
  - Enviromental/Biological Global warming, Climate change, Ozone depletion, Solarflare, Epidemic
- Manmade Disasters
  - Technological (Transport Accidents, Structure failures, Explosions, Fires)
  - Industrial (Chemical spills, Radiation, Poisioning, Gas leaks)
  - Warfare (War, Terrorism, Internal conflicts, Civil unrest)

1) Natural Disasters :- They are the result of naturally occurring processes that have operated throughout Earth's history.

→ Natural disasters are beyond human control.
→ Natural disasters are often termed as "Act of God".

a) Meteorological
→ Events caused by short-lived/small to meso scale atmospheric processes (Minutes to days)

Cylones, Strom

→ Events caused by deviations in normal water cycle/overflow of water bodies

Floods.

b) Geological
→ Events originating from solid earth
Earthquakes, Volcano, Tsunami, Landslides

c) Environmental/Biological
→ Events caused by long-lived/meso to macro scale processes.
Extreme temperature, Global warming

2) Manmade disasters
a) Technological : The failure (or) breakdown of systems, equipment and engineering standards that harms people and environment; structural collapses such as bridges, buildings etc.
b) Industrial : disasters caused by Industrial companies, either by accident, chemical and nuclear explosion.

c) Warfare : Disasters caused by conflicts, wars, intra society conflicts etc.

3) Incidences of Mass Trauma :- Infectious disease outbreaks, incidents of community unrest and other types of traumatic events can also bring out strong emotions in people.

→ Outbreak of Ebola affecting several people in West Africa are examples of Incidences of Mass Trauma.

What are the General Effects of Disaster?

The General effects of disaster may be

→ Loss of life
→ Injury
→ Damage to and destruction of property
→ Damage to cash crops
→ Disruption of production
→ Disruption of lifestyle
→ Loss of livelihood
→ Disruption to essential services
→ Damage to national infrastructure and disruption to government systems
→ National economic loss
→ Psychological effects on people.

Disaster Management (Definition):-

Disaster management is a process of effectively preparing for and responding to disasters.

→ It involves strategeically organizing the resources to lower the harm that disasters cause.

→ It also involves a systematic approach to managing the responsibilities of disaster prevention, preparedness, response and recovery.

Why is Disaster Management Important?

1) To save the life of people
2) Promotes Disease prevention
3) Improves Community Resilience
4) Reduces poverty
5) Improves health
6) Reshapes communities
7) Strengthens security
8) Promotes stability
9) Promotes the protection of Natural Resources
10) Strengthens social contracting and Trust.

1) To save the life of people: Generally lack of coordination and response to disasters can have

serious and long lasting impacts on community and can lead to more deaths.

→ Disaster management can help to enhance the ability of emergency responders to save lives.

2) Promotes Disease Prevention: Disasters cause a large number of deaths, and people are exposed to a range of illness during disasters.

→ Through disaster management practices, communities can improve their health and reduce the diseases.

3) Improves Community Resilience: Training the people about disasters helps to improve the effectiveness of disaster management response teams. which is one of the key elements of disaster management training.

4) Reduces Poverty: Disasters can push people into poverty and change the lives of people.

→ By preparing for disasters, communities can mitigate the threat of poverty, hunger

5) Improves health:- Disaster management team sees that the communities have access to health professionals, have a good supply of water and adequate sanitation facilities to improve the health during disasters.

6) Reshapes Communities: Disaster management can help communities rebuild their communities

and reconnect people with eathother.

7) Strengthens Security: Terrorist group exploit disasters to cause further bloodshed.

→ Through disaster management protection increases and hence strengthens security.

8) Promotes Stability: Disaster can disturb the social order, economic activity and flow of trade.

→ Through proper planning insecurity, hatred and violence can be reduced.

9) Promotes the protection of Natural Resources:-

If a disaster negatively affects the environment it may also lead to species extinctions, destruction of ecosystems.

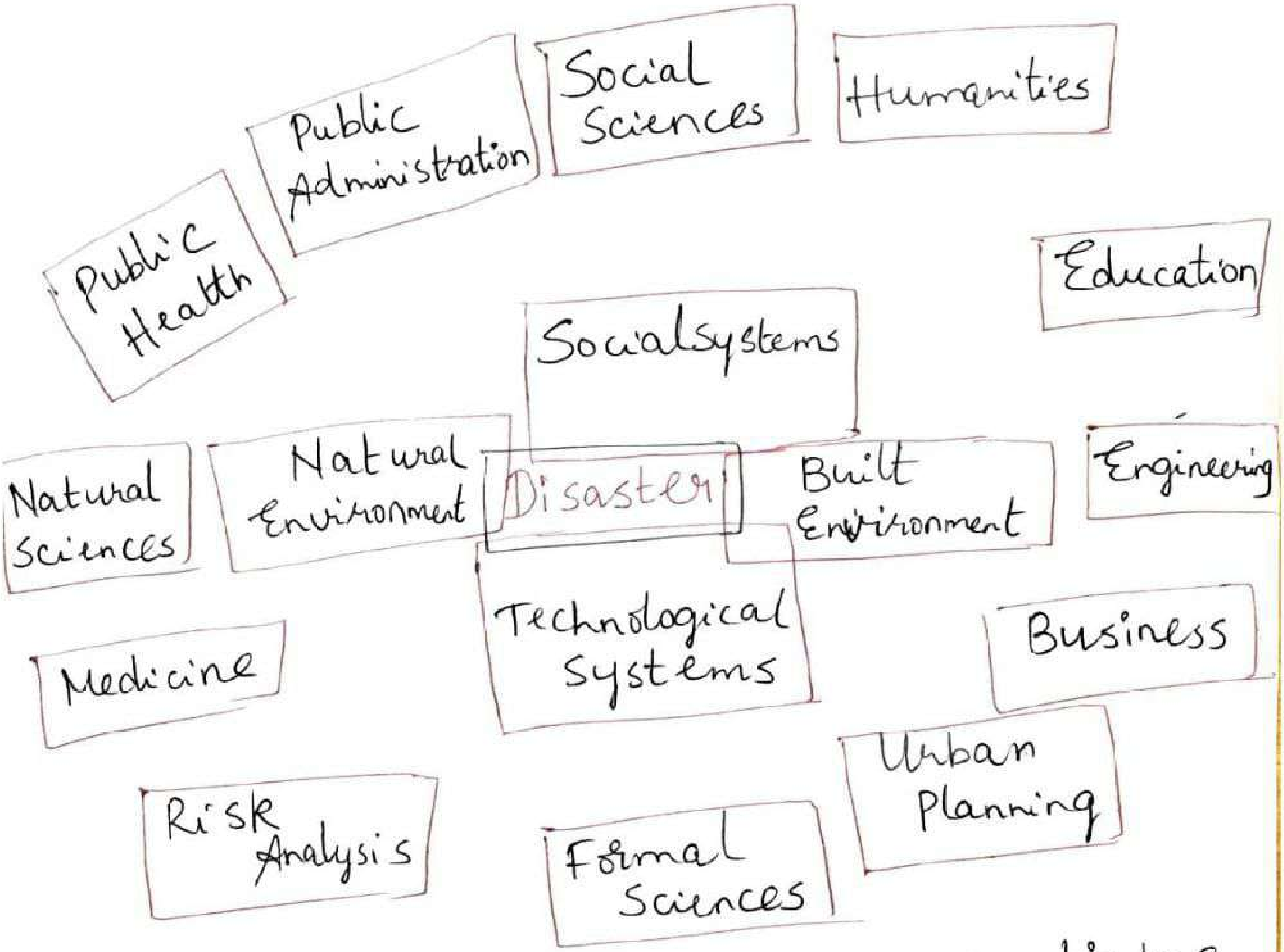
→ communities must plan for disaster and work to secure their natural resources.

10) Strengthens Social contracting and Trust:-

During disasters governments, other important organisations fail to provide people with protection, food, shelter which can lead to social inequalities, distrust etc which may lead to violence activities.

→ Hence by proper preparedness to disasters by disaster management one can strengthen the trust in people.

Inter disciplinary nature of the Disaster Management



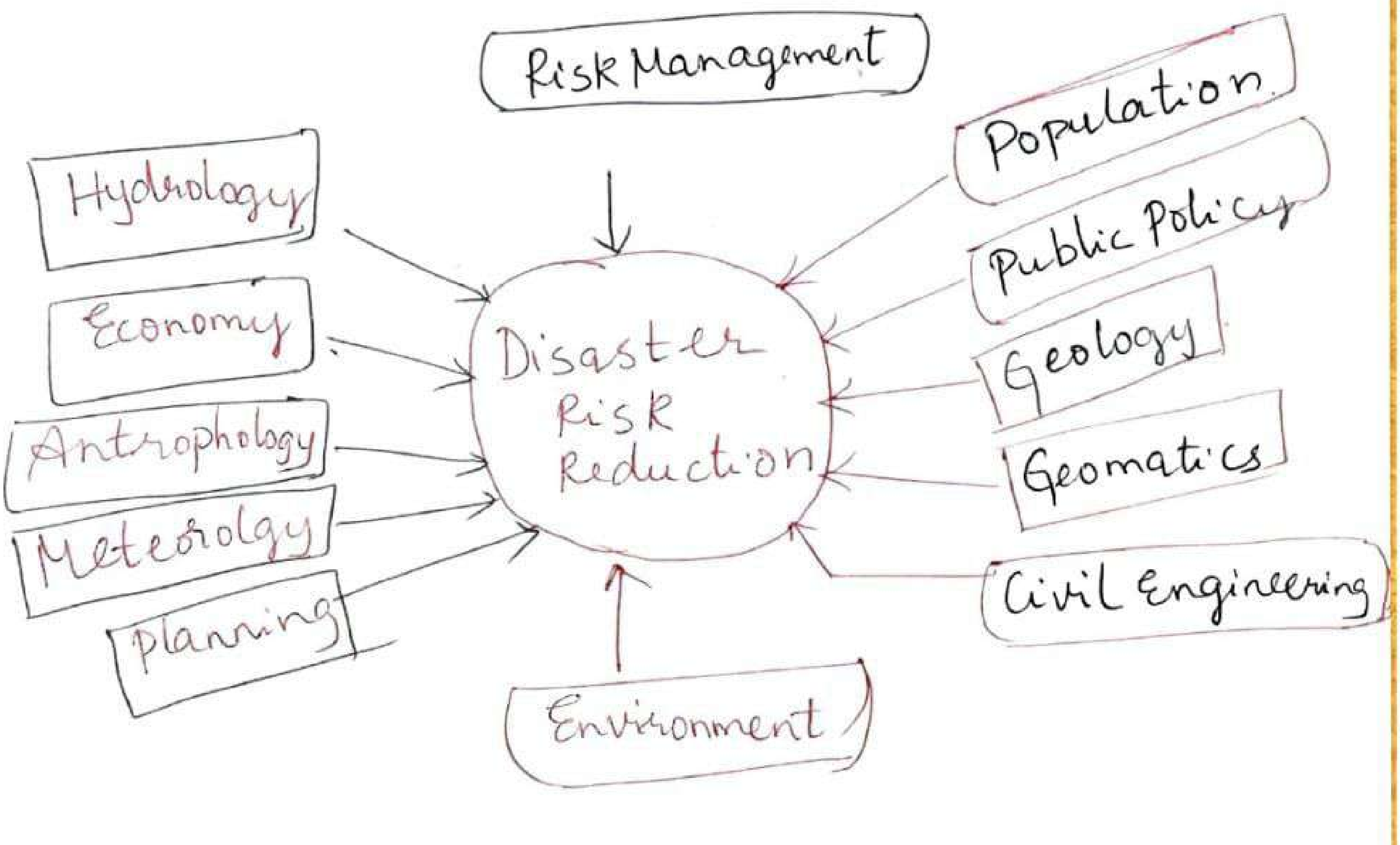
→ Interdisciplinary nature means combining (or) involving two (or) more academic disciplines (or) fields of study

→ Disaster management is how we deal with human, material, economic (or) environmental impacts after the disaster.

→ It is the process of how we prepare for respond to and learn from the effects of major failures.

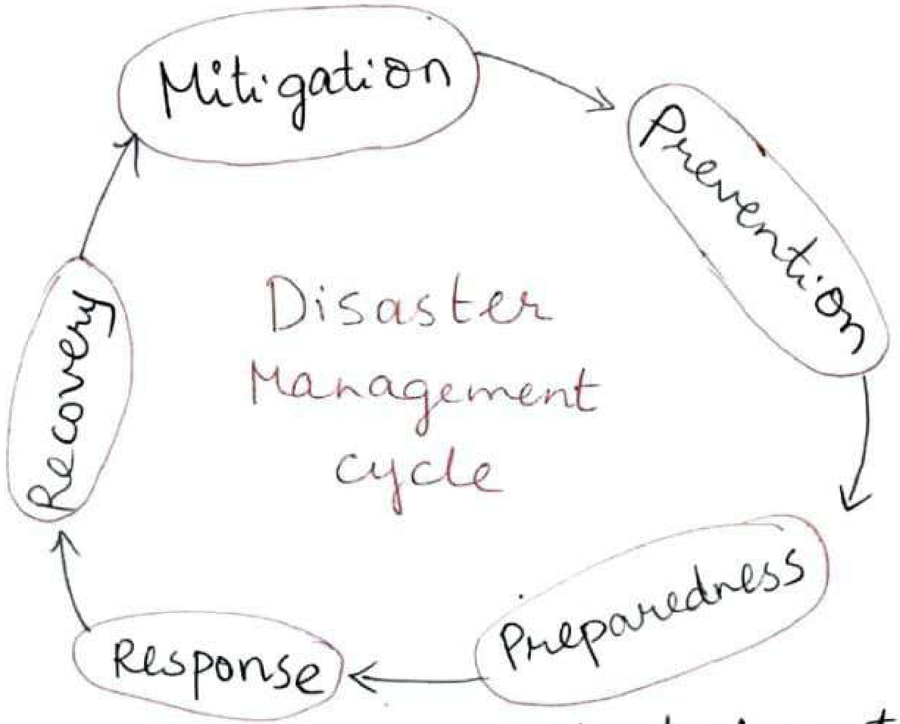
→ As the Disaster management is involved with various disciplines like
- → Hydrology
- → Economy
- → Meteorolgy
- → Planning
- → environment
- → Risk management etc.

It is interdisciplinary in nature.



Interdisciplinary Nature of Disaster Management

Disaster Management cycle – Five priorities for action



→ Disaster management tries to reduce (or) avoid the losses from hazards and provides effective recovery.

→ The disaster management cycle gives the information about the ongoing process by which governments plan for and reduce and the impact of disasters

→ It also deals with how they react during and immediately following a disaster.

The different phases involved in the disaster management cycle are

1) Mitigation

2) Prevention
3) Preparedness
4) Response
5) Recovery.

1) Mitigation: Minimizing the effects of disaster

→ Mitigation activities eliminate (or) reduce the probability of disaster occurrence (or) reduce the effects of unavoidable disasters.

Mitigation measures include

→ Building codes
→ Vulnerability analysis
→ zoing and land use management
→ Following regulations & safety codes.
→ Preventive health care
→ Public education.

2) Prevention: Prevention is to ensure that the human action (or) natural phenomena do not result in disaster (or) emergency.

Prevention measures include

→ Avoid overcrowding
→ Deforestation
→ Following Building codes
→ Public education.

3) Preparedness :- It involves the readiness to respond to any emergency situation through the programs that strengthen the technical and managerial capacity of governments, organizations

Preparedness measures can be

→ Response mechanisms
→ Rehearsals
→ Developing long-term & short term strategies
→ Public education.
→ Storage of food, water, medicines etc

4) Response: The aim of Response is to provide immediate assistance to maintain life, improve health of affected population.

Assistance is provided

→ with transport
→ Temporary shelter
→ Food
→ Establishing semipermanent settlement in camps & locations.

→ The main focus is on meeting the basic needs of the people until the permanent and sustainable solutions can be found.

3) Recovery: As the emergency is brought under control, the affected population is capable of undertaking a growing number of activities aimed at restoring their lives, infrastructure etc.

Recovery measures both short & long term include

→ Returning vital life - support systems to minimum operating standards

→ Temporary housing
→ Public Information.
→ Health and safety education
→ Reconstruction.
→ Counseling programs.
→ Economic Impact studies.

Mitigation ←
↓
Prevention
↓
Preparedness
↓
Response
↓
Recovery →

Disaster Management cycle.

Case Study methods of the following

Vegetal cover floods:

Types of floods :-

1) Flash floods
2) Coastal floods
3) River floods
4) Urban floods
5) Pluvial floods

1) Flash floods :- They are fast moving waters that sweep everything in their path.
→ They are caused by heavy rainfall (or) rapid snow.
→ Floods usually cover a relatively small area and occur with little to no notice, generally less than six hours.

2) Coastal floods :- They are caused by strong winds (or) storms that move towards a coast during high tide.
→ Coastal areas with fewer defenses and lower elevation are most affected.

3) River floods :- They are caused by gradual riverbank overflows caused by extensive rainfall over an extended period of time.
→ The areas covered by river floods depend

on the size of river and amount of rainfall
→ River floods rarely result in loss of lives but can cause immense economic damage

4) Urban floods: They occur when the drainage system in a city (or) town fails to absorb the water from heavy rain.
→ The lack of natural drainage in urban area can also result in flooding.
→ It can cause significant structural damage.

5) Pluvial floods:- They occur in flat areas where the terrain can't absorb the rain water, causing puddles and pond's to appear.
→ Pluvial flooding is similar to urban flooding, but it occurs in rural areas.

Effects of floods:

→ Every year millions of people become homeless, for many days due to floods.
→ Millions of houses are damaged and large no. of them will be collapsed.
→ Millions hectares of agricultural land come under the deep flood water and not in the condition for further cultivation
→ Thousand of hectares of land have been converted into waste land / barren land.
→ Production of certain agricultural crops including cash crops are gone down.

→ Due to the large quantities of flood water at certain places for long time various types of water borne diseases suddenly rise up
→ National and state highway's including other associated link roads have been submerged in flood water.
→ The power supply has been totally damaged both with water and electricity because almost all electric poles are uprooted.

*Flood disaster Management (or) Mitigation Measures
The various measures adopted for flood disaster management may be divided into two groups.

1) Struictural
2) Non structural.

Structural measures may be grouped into the following

→ Dams & Reservoirs
→ Embankments, Flood walls, seawall
→ Channel Improvement
→ Drainage Improvement
→ Diversion of flood waters

→ For the effective functioning of all the physical measures taken, it is necessary that pre and post monsoon checks must be made & repairs are to be made if any.

The various non-structural measures implement-ed are

1) Modifying the susceptibility to flood damages through.

1) Flood plain management

2) Flood proofing including disaster preparedness and response planning

3) Flood forecasting & Warning

1) Flood plain management:

→ The basic concept is to regulate the landuse in flood plain zoning to reduce damage due to floods.

→ Zoning is done by determing the locations, extent of areas which are affected by floods of different magnitudes.

2) Flood Proofing:

→ A programme for flood proofing provides the raised platforms for flood shelter for men & cattle.

3) Flood forecasting & Warning

→ This programme was first started in India in 1958 for rive Yamuna.

→ By forecasting the floods we can alert the people.

2) Modifying the flood loss burden through

→ disaster relief.

→ Public Health measures.

* Floods in Coastal Areas in Andhra Pradesh

→ Vast areas in Srikakulam, Visakapatnam districts wer affected by floods in the Vamsadhara and Nagavali and torrential rain in north coastal Andhra on Thursday 10th August 2006 in the wake of deep depression that crossed Orissa coast on 10th August 2006.

→ Flights from Visakapatnam were cancelled for the day as airport came under a sheet of water.

→ Srikakulam district received 18 cm of rain on 10th August 2006.

→ Vamsadhara and Nagavali rivers rose due to huge inflows from Orissa

→ Nearly 300 villages in 19 mandals were partially (or) fully under water.

→ Vizianagaram town experienced 20cm of rain

→ At least 40 persons died in rain and floods in hundred of villages in Srikakulam, Vizianagaram

→ Several localities in the town were Knee deep water.

→ An alarming situation prevailed in Khammam district as the Godavari crossed the danger level of 53 feet in the evening at Bhadrachalam

→ Flights were suspended from Visakhapatnam for the 2nd day also.

Droughts: A drought is defined as a period of abnormally dry weather sufficiently prolonged from the lack of water to cause serious hydrologic imbalance in the affected area.

→ Any area persistently receiving less than 40% of its average rainfall quantity is termed as drought area.

Types of Droughts:-

1) Meteorological Drought
2) Hydrological Drought
3) Agricultural Drought
4) Socioeconomic Drought.

1) Meteorological Drought: It is based on the degree of dryness (or) rainfall deficit and the length of the dry period.

2) Hydrological Drought: It is based on the impact of rainfall deficits on the water supply such as stream flow, reservoir, lake levels and ground water decline

3) Agricultural Drought: It refers to the impacts on agriculture by factors such as rainfall deficits, reduced ground water etc

4) Socioeconomic Drought: It considers the impact of drought conditions (meteorological, agricultural (or) hydrological drought) on

supply and demand of some economic goods such as fruits, vegetables, grains etc.

* Mitigation Measures of droughts

Pre-disaster Response: It will forecast the approaching calamity and give a warning to government agencies to start emergency process.

During disaster and post disaster respone:
The following agencies will take the responsibility to provide the facilities.
→ Central & state government agencies.
→ World Food organization.
→ Non Government Organizations (NGO's)

Short term drought management strategies
→ Supplementary feeding schemes for children under the age of 5
→ Free food programmes, ment to meet the requirements of elderly, disabled people in drought affected areas
→ Food for work programmes
→ Grain loan scheme.

Long term drought management strategies
→ Research into drought resistant crops as well

as food storage at the house hold level
→ Promotion of the agricultural growth in the semi-arid areas
→ Establishment of seed banks to see that the seeds are available after drought
→ Promotion of alternate cattle food.
→ Agricultural loans to the affected people

Effects of Drought:

1) Hunger and Famine: Drought conditions often provide too little water to support food crops hence people go hungry due to lack of sufficient food materials
2) Not enough drinking water: All living things must have water to survive. During drought there will be no enough drinking water.
3) Diseases:- Drought often creates a lack of clean water for drinking, public sanitation and personal hygiene which can lead to spread of various diseases
4) Wild fires:- The low moisture and precipitation that is characterized due to droughts can quickly result in forest fires.
5) Wild life: Wild plants & animals suffer from droughts due to lack of water
6) Electricity Generation: Drought will reduce the amount of water & hence reducing the amount of power generation.

* Earthquakes

Earthquake (Definition): Earthquake is a term used to describe both sudden slips on a fault (break in continuity of rock strata) and the resulting ground shaking and radiated seismic energy caused by the slip, or by volcanic (or) other sudden stress changes in the earth.

* Types of Earthquake

There are four types of earthquakes

1) Tectonic Earthquakes
2) Volcanic Earthquakes
3) Collapse Earthquakes
4) Explosion Earthquakes

1) Tectonic Earthquakes:

→ They are most popular earthquakes observed all over the world, these are generated due to the sliding of rocks along a fault plane

→ It is generated by the movement of plates when energy accumulated within plate boundary zones is discharged

→ Tectonic earthquakes are normally larger than volcanic earthquakes

2) Volcanic Earthquakes :- An exceptional class of tectonic earthquakes are recognized as Volcanic earthquakes.
→ These are restricted to areas of active volcanoes.
→ These are less devastating when compared to remaining.

3) Collapse Earthquakes
→ In the areas of intense mining activity, sometimes the roofs of underground mines collapse generating minor temors.
→ These are called as Collapse Earthquakes.
→ Collapse earthquakes are also produced by massive land sliding

4) Explosion Earthquakes:
→ Ground shaking may also happen due to the explosion of chemical (or) nuclear devices.
→ Such tremors are called explosion earthquakes.
→ Some underground nuclear explosions fired since 1950's produced explosion earthquakes.
→ When a nuclear device is detonated in a bore hole underground a large amount of nuclear energy is released.

* Effects of Earthquakes

1) Shaking and Ground Rupture:
→ It is the first outcome generated by the earthquake mainly resulting in more/less severe loss to structures
2) Soil liquefaction: It occurs when water saturated granular material (such as sand) temporarily loses its strength and transforms from solid to a liquid because of the shaking
3) Human Impact: An earthquake may lead to injury and loss of life, general property damage, road & bridge damage etc.
4) Landslides: Earthquakes can generate slope instability driving to land slides, a major geological hazard.
5) Fires:- Earthquakes can also lead to fires by destroying electrical power (or). gas lines.
6) Tsunami: Tsunami's are long-wavelength, long period sea waves originated by the sudden (or) abrupt movement of large volume of water.
7) Floods:- Floods can also be secondary impacts of earthquakes if dams are damaged

* Causes of Earthquake

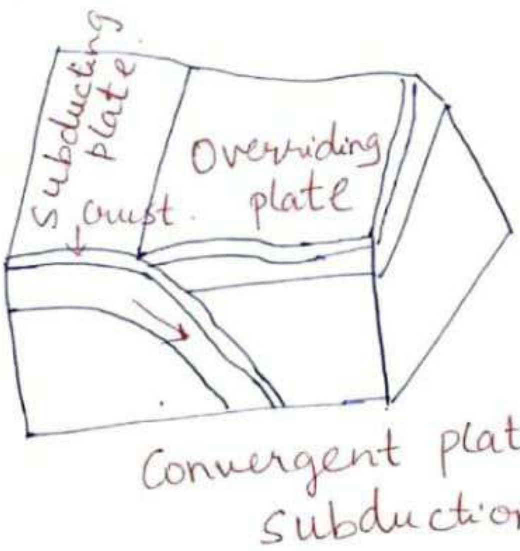
→ The earth's crust is a rocky layer of varying thickness ranging from a depth of about 10 Km under the sea to 65 Kms under the continents.

→ The crust is not one piece but consists of portions called 'plates' which vary in size.

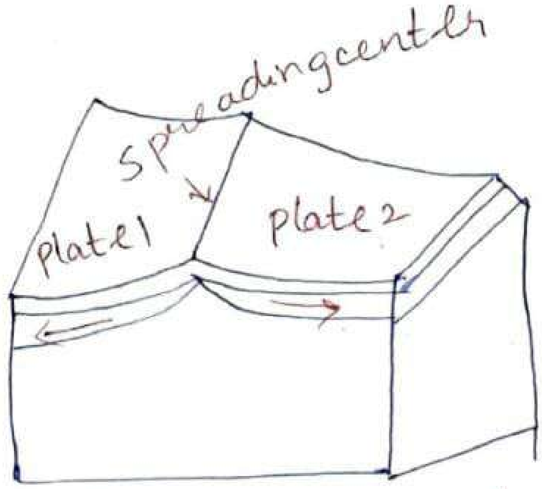
→ When these plates contact each other, stress aries in the crust.

→ These stresses can be classified as per the type of movement along the plate boundaries.

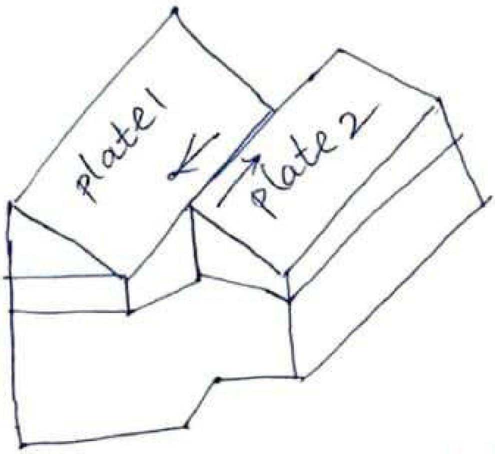
a) Pulling away from each other
b) Pushing against one another
c). Sliding sideways relative to each other.



Convergent plate bounding
subduction zone.



Divergent plate boundary



Transform plateboundary

- Earthquakes can be of three types based on the focal depth

(1) Deep : 300 to 700 kms from earth surface

2) Medium : 60 to 300 kms

3) Shallow : less than 60 kms

→ The deep focus earthquakes are rarely destructive because by the time the waves reach the surface the impact reduces.

* Preventive Measures for Earthquakes / Mitigation measures:-

1) Community preparedness:- Community preparedness is important for minimising the impact of the earthquakes
→ The most effective way to save you even in a slightest shaking is DROP, COVER & HOLD.

2) Planning: The Bureau of Indian standards has published building codes and guidelines for safe construction of buildings against earthquakes
→ Before the buildings are constructed the building plans must be checked by municipality.

3) Public Education: Public education is educating the public on causes of earthquake & its preparedness measures.

4) Engineered structures:- Buildings need to be designed and constructed as per the building by laws to withstand ground shaking
→ Architectural and engineering inputs are to be put together to improve building design.
→ Buildings on soft soil are more likely to get damaged even if the magnitude of earthquake is low.

* Land slides:-

Land slide (definition):- The sliding down of a mass of earth, rock etc from the side of a cliff, hillside (or) cutting is called a landslide.

* Causes of Land slides

1) Geological weak material: Weakness in the composition and structure of rock (or) soil may cause land slides
2) Erosion: Erosion of slope due to cutting down of vegetation, construction of roads etc
3) Intense rainfall: Storms that produce intense rainfall for long time can cause land slides
4) Earthquake shaking: Earthquakes can generate slope instability which results in landslides a major geological hazard.
5) Volcanic eruption: Due to volcanic eruption also landslides can occur. as the loose volcanic ash is deposited on hill slides
6) Human Excavation: Excavation of slope, Mining, deforestation, irrigation, vibration, blast etc all the human actions leads to the cause of land slides.

* Preventive measures / Mitigation Measures of land slides

1) Retaining walls: These can be built to stop land from slipping (these walls are commonly seen along roads in hill stations)
→ These are constructed to prevent smaller sized and secondary land slides.

2) Surface Drainage control works: The surface drainage control works are implemented to control the movement of land slides

3) Engineered structures: The structures with strong foundations can withstand (or) take the ground movement forces.
→ Underground installations (pipes, cables etc) should be made flexible to move inorder to withstand forces caused by land slides.

4) Increasing Vegetation cover: It is the cheapest and most effective way of controlling land slides
→ This helps to bind the top layer of soil with layer below

5) Land use practices:
→ No construction of buildings in areas beyond a certain degree of slope.
→ In construction of roads, canals etc care must be taken to avoid blockage of natural drainage.

* Global Warming

Global warming : It is a term used to describe an increase of the average temperature of Earth's atmosphere and oceans.

* Causes of Global Warming

Man-made causes

1) Deforestation: Plants are the main cause/source of oxygen.
→ They take in CO_2 and release oxygen. maintaining environmental balance.
→ Forests are depleted for many domestic and commercial purposes.
→ The above circumstances will lead to global warming
2) Use of Vehicles : The use of vehicles, even for a very short distances results in gaseous emissions.
→ Vehicles burn fossil fuels which emit large amount of carbondioxide resulting in temperature increase.
3) Industrial Development : By the rapid increase in Industries, the temperature of the earth has been increasing.
→ The harmful emissions from the factories add to the increasing temperature of the earth.
4) Agriculture :- Various farming activities produce carbondioxide & methane gas. & increase the temperature of earth

Natural causes of Global warming

1) Valcanoes:- Valcanoes are one of the largest natural contributors to global warming
→ The ash and smoke emitted during volcanic eruptions increases the temperature.

2) Water Vapour: Water Vapour is a Kind of green house gas.
→ Due to the increase in earth's temperature, more water gets evaporated from water bodies & adds to global warming.

* Effects of Global Warming

1) Rise in Temperature: Global warming has led to an incredible increase in earth's temperature.
→ this has resulted in an increase in the melting of glaciers which led to an increase in the sea level

2) Threats to the Eco system: Global warming has affected the ecosystem that can lead to the loss of plant & animal lives due to rise in the temperatures.

3) Climate change:- Global warming has led to a change in climatic conditions
→ There are droughts at some places and floods at some
→ this climatic imbalance is result of global warming

4) Spread of Diseases:- Global warming leads to change in the patterns of heat and humidity

5) High Mortality Rates:- Due to an increase in floods, tsunami's & other natural calamities, the average death rate usually increases

6) Loss of Natural Habitat: A global shift in climate leads to loss of habitats of several plants and animals

Mitigation Measures / Preventive measures of Global Warming

1) Establish Early Warning system: By giving the early warning about Global warming to the residents they take proper precautions to reduce the effects of global warming.

2) Capacity building / Traning programme:- It is to be done for health care professionals at local level to recognize and respond to heat releated illness, due to extreme heat waves.

→ These traning programs should focus on medical officers & community health staff to reduce mortality.

3) Public Awareness and Community outreach:-
Spreading public awareness messages on how to protect against the extreme heat waves are one of the important measure to reduce (or) minimise the effects of global warming.

4) Collaboration with non government and civil society:- Collaboration with non-government organizations, civil society organizations to improve bus stands, building temporary shelters reduces the effect of heat waves.

* Cyclones:

Cyclone (Definition):- Cyclone is a region of low atmospheric pressure surrounded by high atmospheric pressure resulting in swirling atmospheric disturbance accompained by powerful winds blowing in anticlockwise in the Nothern Hemisphere and in clockwise direction in the Southern Hemisphere.

cyclones are known by different names in different parts of world.

→ Typhoons - North west Pacific ocean
→ Hurricanes - North Atlantic ocean
→ Tropical cyclones - Southwest pacific ocean
→ Severe cyclonic storm - North Indian ocean
→ Tropical cycle - South west Indian ocean
→ Willie - Willie in Australia
→ Tornada in south America.

* Effects of Cyclones:-

→ High winds cause major damge to infrastructur and housing

→ They are generally followed by heavy rains and floods.

1) Physical damage: Structure will be damaged (or) destroyed by the wind force, flooding and storm.

→ Light pitched roofs of most structures especially the ones fitted on to industrial buildings will suffer severe damage.

2) Public Health:- Public health has serious effects due to flooding, contamination of water supplies, diarrhoea, malaria etc

3) Water supplies:- Ground and pipe water supply may get contaminated by flood waters and food supplies - high winds and rain

→ Crops and food supplies - high winds and rain disturbs/damages the standing crop and also food stock lying in low lying areas

→ the loss of crop may lead to food shortage

4) Communication:- Severe disruption in the communication links as the wind may bring down the electricity and communication towers, telephone poles, telephone lines, antennas etc

→ Lack of proper communication affects the distribution of relief materials.

*Risk Reduction Measures / Mitigation Measures of cyclones

Coastal Belt Plantation:- Green belt plantation along the coastal line in scientific interweaving pattern can reduce the effect of hazard.
→ Forests act as a wide buffer zone against strong winds and flash floods.
Hazard Mapping: Meteorological records of the wind speed and the directions give the probability of winds in the region.
→ cyclones can be predicted several days in advance
→ A hazard map will give the information of areas affected by cyclones every year.
Land use control: It is designed so that least critical activities are placed in vulnerable areas.
→ Policies should be in place to regulate land use and building codes should be enforced.
Engineered structures: → structures need to be built to withstand wind forces.
→ Houses can be strengthened to resist wind & flood damage.
→ A row of planted trees will act as shield
→ Communication lines should be installed underground
Flood Management: Torrential rains, strong wind leads to floods in cyclone affected areas.
Improving Vegetation cover: The roots of the plants & trees keep the soil intact and prevent erosion & slow runoff to prevent (or) lessen flooding.

* Tsunami

Tsunami (Definition):- The term Tsunami has been derived from a Japaneese term Tsu meaning 'harbor' and nami meaning 'waves'.

→ Tsunamis are popularly called tidal waves.

→ These waves often affect distant shores, originate by rapid displacement of water from sea either by seismic activity, landslides etc.

* Causes of Tsunami:

1) Earthquake: The most common are the fault movements on the sea floor accompained by an earthquake

→ They release huge amount of energy and have capacity to cross oceans.

2) Landslides: The second most common cause of the tsunami is a landslide either occuring under water (or) originating above the sea.

3) Valcano: The third major cause of tsunami is volcanic activity.

→ If valcano is located near to shoore (or) under water may be uplited (or) depressed similar to the action of fault (or) the volcano may explode

the above all are the causes of the Tsunami

Possible risk reduction Measures / Mitigation measures

Site Planning & Land Management:- It determines the location, configuration and density of development on particular site and hence helps in reducing tsunami risk.

→ This strategy is designed to keep development at a minimum in hazard areas.

Engineering structures:

→ Site selection - Avoid building (or) living in buildings within several hundered feet of coastline

→ Construct the structure on a higher ground level with respect to mean sea level.

→ Construction of water breakers to reduce the velocity of waves.

→ Use of water and corrosion resistant materials for construction.

→ Construction of community halls at higher locations, which can act as shelters at the time of disaster.

Flood Management:

Flooding will result from a tsunami.

→ Tsunami waves will flood the coastal areas

→ Flood mitigation measures must be taken.

* Adverse effects | Post Hazards of Tsunami.

→ Local tsunami events (or) those less than 30 minutes from the source cause the majority of damage

→ It is normally the flooding affect of the tsunami that causes major destruction to the human settlements, roads, infrastructure etc

→ As the waves withdraw towards the ocean they sweep out the foundations of the buildings, the beaches get destroyed.

→ Damage to ports and airports

→ There is huge impact on public health system

→ Increase in death rate.

→ sewage pipes may be damaged causing major sewage disposal problems

→ Availability of drinking water has always been major problem in areas affected by disaster.

→ Open wells and other ground water may be contaminated by waste & sewage

→ Flooding in the locality may lead to crop loss, loss of livelihood etc.

→ Total functioning of human beings will be affected.

Five Priorties for Action in DM

The Hyogo framework for Action (HFA) was adopted by the United Nations to reduce disaster losses: It identified five priorties for actions to make nations and Communities resilient to disasters.

life safety

Priority 1:- Ensure that disaster risk reduction is National & local priority

→ DM should be integrated into national & local development policies.
→ strong institutional frameworks & legislation are essential.
→ Adequate funding & Coordination among agencies are required.
→ Establish DM authorities.
→ Allocate budget for disaster preparedness.
→ strengthen goverance & Coordination

Communication

Priority 2:- Identify, Assess and Monitor Disaster Risks & enhance Early Warning

→ Understanding hazards, vulnerabilities and risks is crucial.
→ Use of risk assessment tools, hazard mapping & early warning System
→ Timely dissemination of warnings to the public.
→ Hazard mapping & risk analysis
→ Use of GIS & RS.
→ Early warning systems for floods, cyclones, earthquakes etc.

Damage assessment

Priority 3:- Use Knowledge, Innovation and Education to build a culture of safety & Resilience.

→ Public awareness & education reduce disaster impacts
→ Incorporation of DM in school & college curricula.
→ Sharing best practices & research findings.

→ Disaster education & training programs.
→ Community awareness Campaigns.
→ Capacity building through innovation & technology.

Recovery & restoration

Priority 4 :- Reduce the Underlying Risk factors.

→ Address root Causes of disasters such as poor land use planning, environmental degradation & unsafe construction.
→ Promote Sustainable development practices
→ Enforce building Codes & standards.
→ Protect ecosystems
→ Improve Urban planning
→ Reduce poverty & social vulnerability.

Preparedness & prevention.

Priority 5 :- Strengthen Disaster Preparedness for effective response at all levels.

→ Preparedness ensures quick & effective response during disasters.
→ Development of emergency response plans, mock drills, resource mobilization.
→ Disaster preparedness plans
→ Training of response teams
→ Emergency shelters & evacuation plans
→ Regular mock drills.